Obesity:

Epidemic, Complications and Comorbidities, Impact of Weight loss

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Learning Objectives

- Discuss knowledge of obesity-related complications and comorbidities and identify the benefits of weight reduction.
- Discuss the trends in the epidemiology of Obesity
- Describe the variations in obesity trends between high income and low to middle income countries.

Recap: Introduction to Obesity

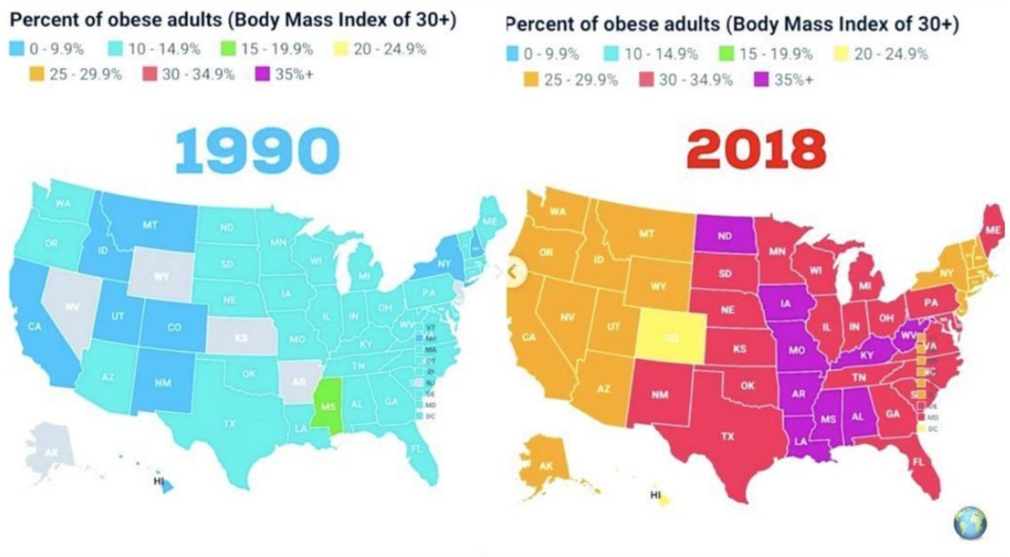
- In Semester 3 CARD Module, you were introduced to the concept of obesity as a disease.
- The American Medical Association (AMA) designated obesity a disease in 2013.
- Definition:

  Obesity is a "chronic, relapsing, multi-factorial, neurobehavioral disease, wherein an increase in body fat promotes adipose tissue dysfunction and abnormal fat mass physical forces, resulting in adverse metabolic, biomechanical, and psychosocial health consequences.

Why do you think it is important to study the progression of the obesity epidemic?

Understanding the Obesity Epidemic

Studying the obesity epidemic and its variations by age, gender, socioeconomic status and countries helps us to unravel the driving forces for the perpetuation of the epidemic and it can be useful in informing the development of public health interventions.



Global Obesity Prevalence and Projections

The worldwide prevalence of obesity has more than tripled between 1975 and 2022. Obesity is now recognized as one of the most important public health problems facing the world today.

2020

2.2 Billion

persons who are overweight/obese

44% Prevalence

2035

3.3 Billion

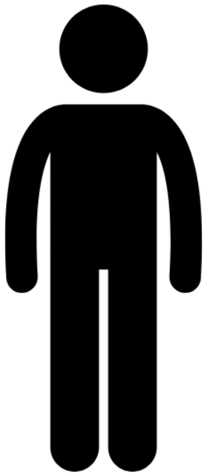
persons who are overweight/obese

54% Prevalence

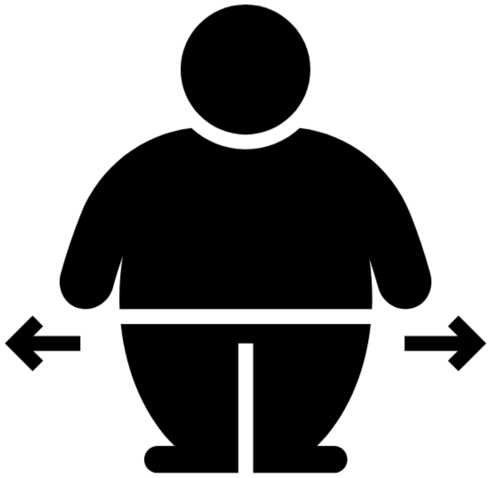
Table 1.1: Global estimate (2020) and projected number of adults (2025-2035) with high BMI

	2020	2025	2030	2035
Adults with overweight (BMI ≥25 to 30 kg/m^2)	1.39bn	1.52bn	1.65bn	1.77bn
Adults with obesity (BMI ≥30 kg/m^2)	0.81bn	1.01bn	1.25bn	1.53bn
Adults with overweight or obesity as a proportion of all adults globally	42%	46%	50%	54%

Source: World Obesity Federation, 2023b



Global age-standardized mean BMI for men



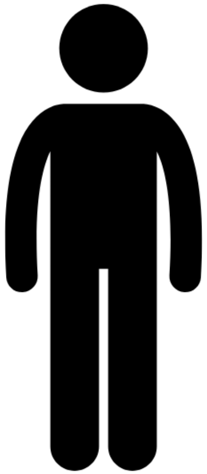
21.7 kg/m2

1975

24.2 kg/m2

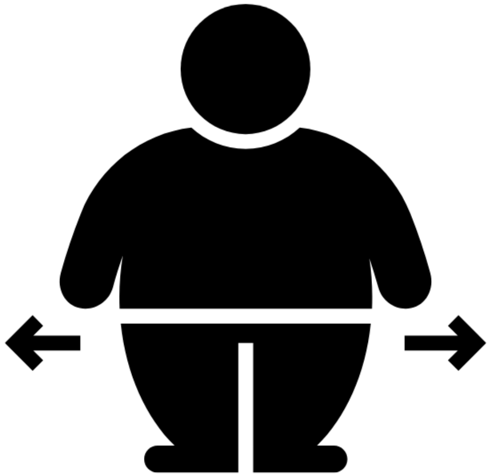
2014

Global age-standardized mean BMI for women



22.1 kg/m2

1975



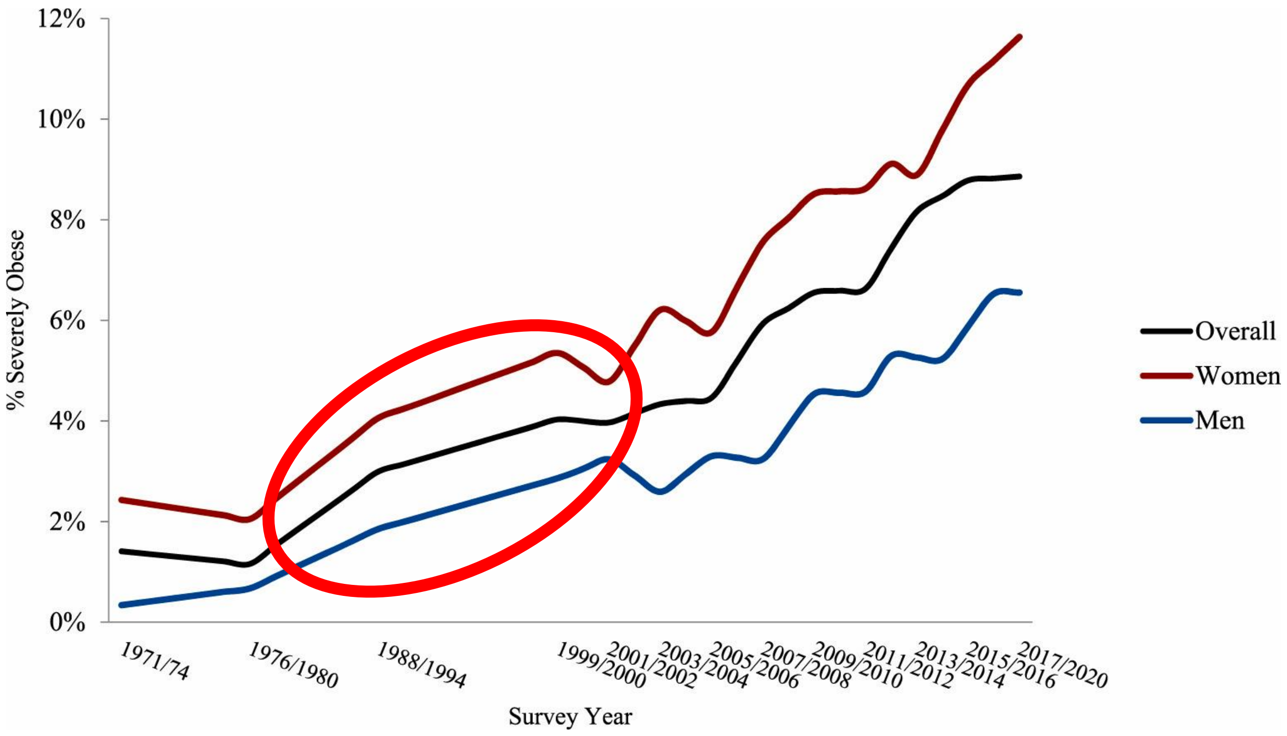
24.4 kg/m2

2014

Explosive Rise in 1970 to 1980s

The rising tide of the obesity epidemic began almost simultaneously in most developed countries in the 1970s and 1980s.

One theory that has been posited to explain this phenomenon is the "Energy Balance Flipping Point Hypothesis" – our first clue into the drivers of the obesity epidemic!



Historical trends in severe obesity for adults age 20+ years, 1971–2020: NHANES

Energy Balance Flipping Point Hypothesis

According to the energy flipping point hypothesis, there was a critical switch in energy balance in the 1960s and 1970s.

This critical point, called the "flipping point" divides the timeline of the obesity epidemic into two distinct phases:

- PULL PHASE
- PUSH PHASE

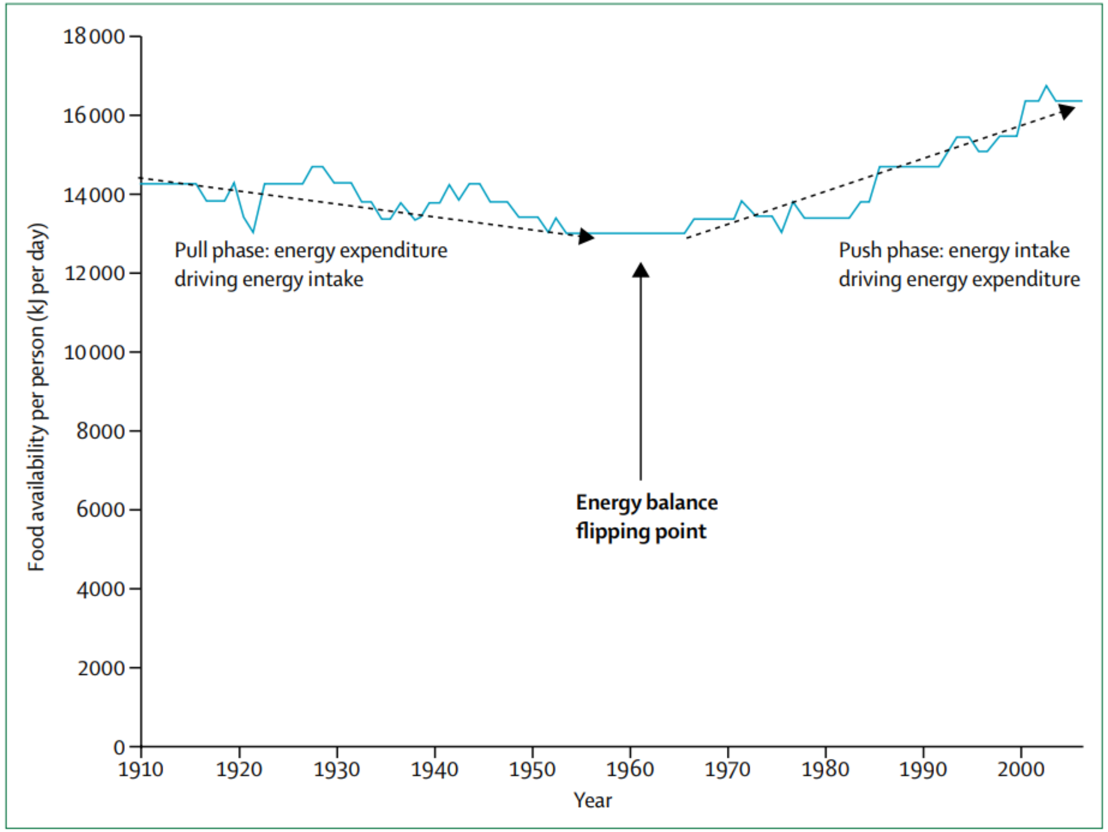


***Figure 3*: Food availability for the USA, 1910–2006**[58]
There are two distinct phases: a decrease in food energy supply (postulated to be pulled down by reduced energy expenditure requirements for daily living), followed by an increase in food energy supply (postulated to be pushed up by increasing food access). An energy balance flipping point is proposed, marking the change in how the US

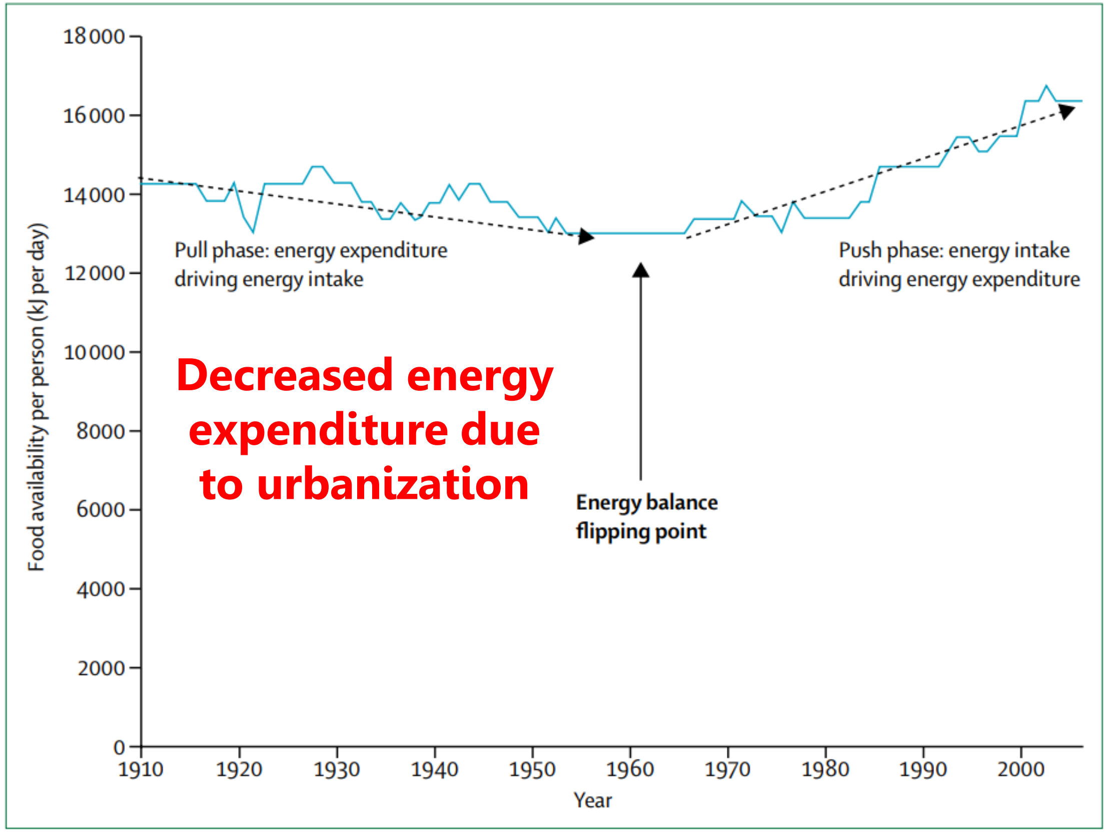
Flipping Point Hypothesis: Pull Phase

Also called the "move less, stay lean" phase.

The pull phase is the first half of the twentieth century. (1910 – 1960)

There was increasing urbanization and mechanization leading to reduced energy expenditure requirements for daily living.

This reduced energy expenditure "pulled" down energy intake, so that persons also ate less and thus their weight was stable and obesity prevalence was relatively stable in the "pull" phase.



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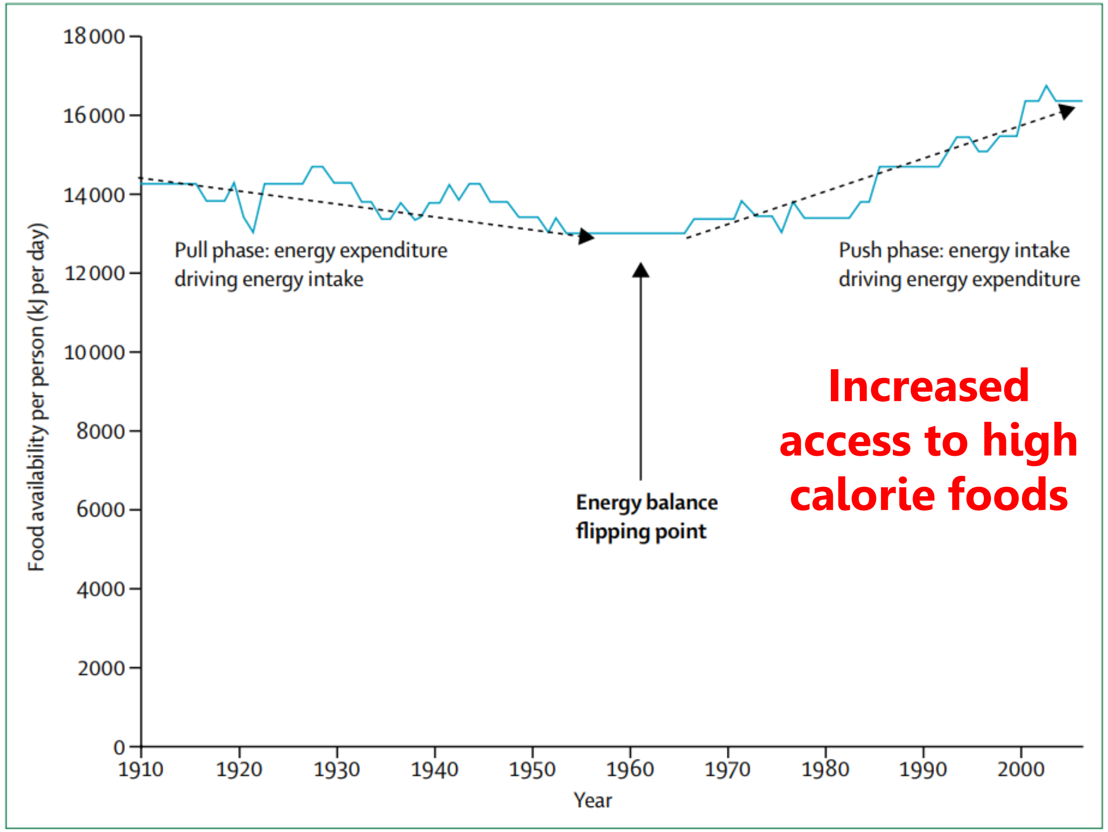
Flipping Point Hypothesis: Push Phase

Also called the "eat more, gain weight phase" phase.

The pull phase is the first half of the twentieth century. (post 1960s)

Persons now have plentiful access to cheap nutrient void, calorie dense foods. This led to increased energy intake while persons continued to move less. This resulted in an explosive increase in obesity prevalence in the post 1960s



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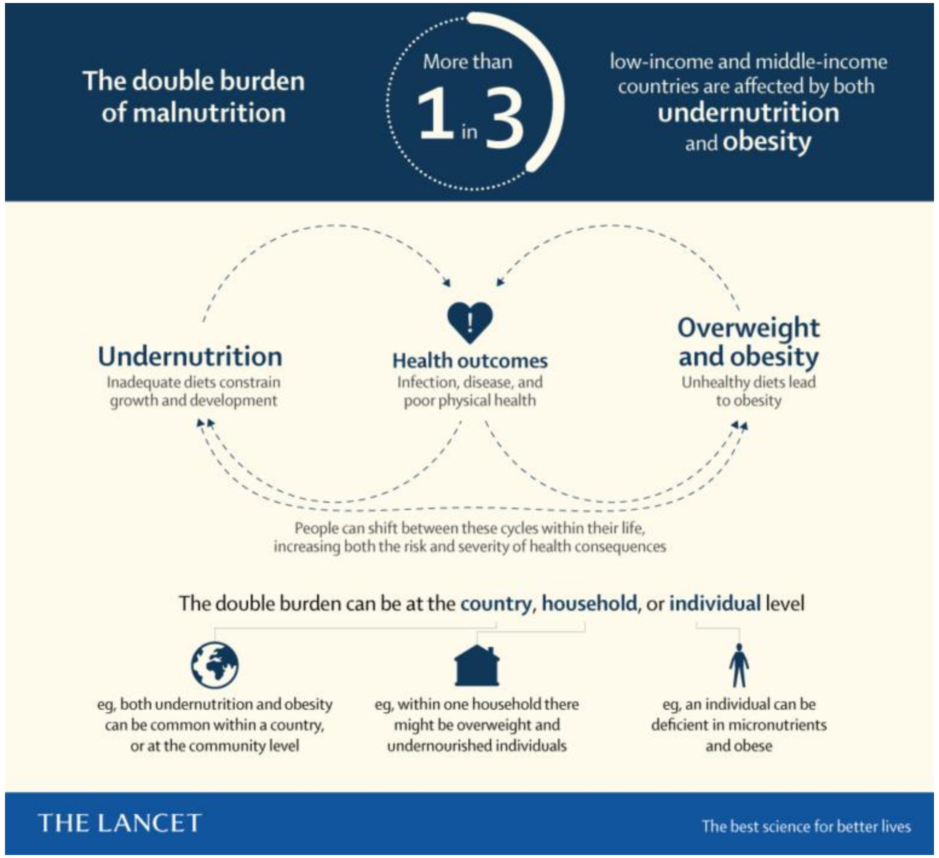
Global Variations in Obesity Trends

- The socioeconomic distribution of this increase seems counterintuitive, with 55% of the global increase in obesity occurring in the low-income, WHO African Region.

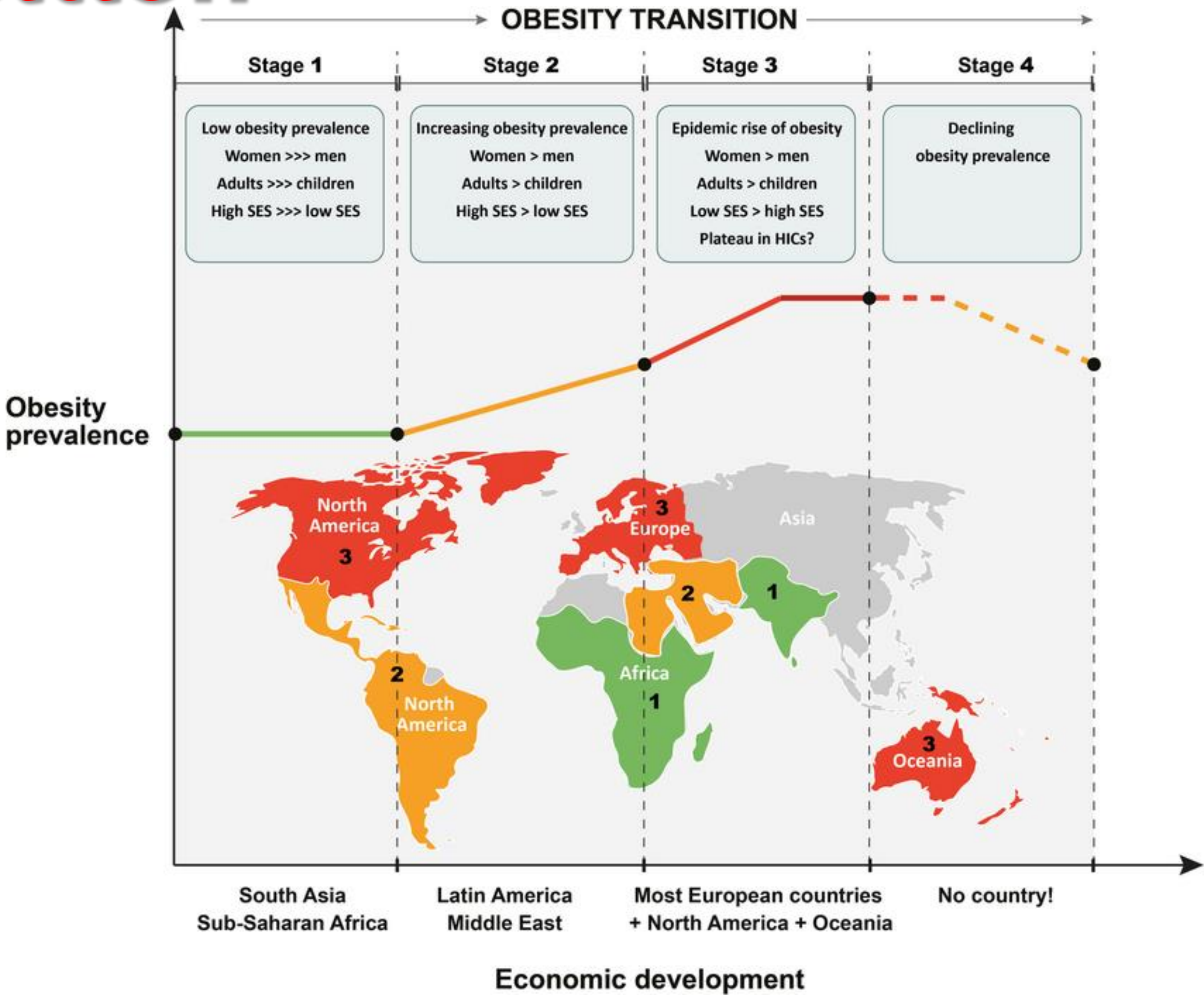
- This highlights the growing phenomenon of the double burden of malnutrition, defined as concurrent consequences of over and undernutrition, occurring in many low and middle-income countries (LMIC).



The “Obesity Transition” Theory

The “Obesity Transition” theory describes how the epidemiology of obesity changes as countries progress economically and become more urbanized.

According to this theory, countries progress through four (4) distinct stages.



Impact of Socioeconomic Status

The influence of socioeconomic status varies both between countries and within countries based on their GDP.

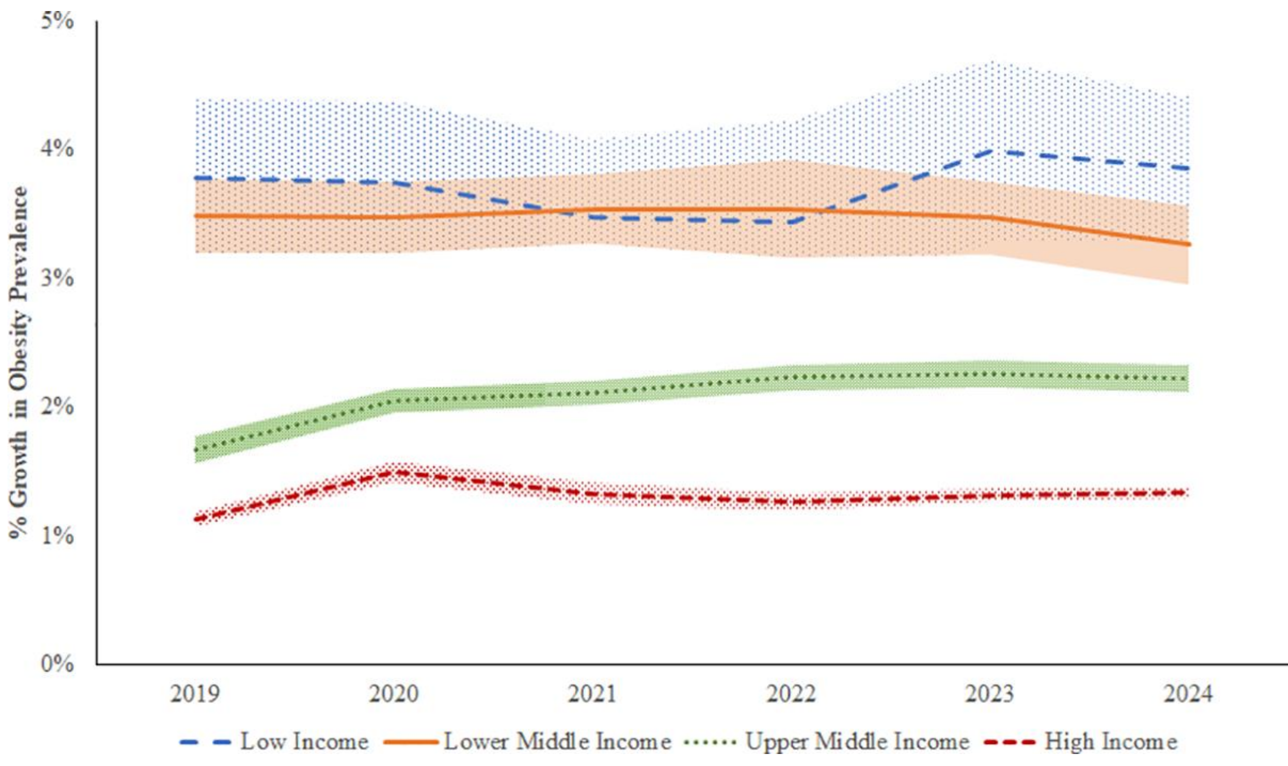
In low-income countries (LMICs), obesity primarily affects middle-aged, wealthy persons, especially women from urban areas.

In Brazil, one of the few middle-income countries with multiple cross-sectional surveys on BMI, this trend was especially noticeable among women, with obesity rates rising swiftly in the lowest income groups.

Alternatively, in high-income countries (HICs), obesity affects both sexes and all age groups but with a disproportionately greater impact in lower socioeconomic strata.

Global Variations in Obesity Trends

The relationship between income and obesity appears contradictory: obesity increases as low-income countries become wealthier, but in high-income countries, it's more prevalent among the less affluent. This suggests a positive correlation in poorer nations and a negative one in richer nations.



Factors Influencing the Obesity Epidemic in LMICs

Table 1.2: Top 20 countries for the highest proportion of adult men and women living with high BMI 2020

	Proportion of men with high BMI		Proportion of women with high BMI
Tonga	80%	Tonga	87%
Samoa	79%	Samoa	86%
United States	79%	Kuwait	79%
Malta	78%	Jordan	78%
Kuwait	77%	Saudi Arabia	78%
New Zealand	76%	Qatar	77%
Australia	76%	Turkey	76%
Israel	76%	Libya	75%

High Income Countries:

- **United States**
- **Malta**
- **Kuwait**
- **New Zealand**
- **Australia**
- **Isreal**

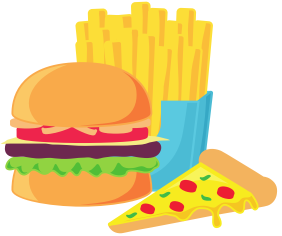
Factors Influencing the Obesity Epidemic in LMICs

Table 1.3: Top 20 countries for the most rapid increase in the proportion of adults living with high BMI 2000-2016

	Compound annual growth in adult obesity 2000-2016 (%)
Lao People's Democratic Republic	3.8
Viet Nam	3.8
Maldives	3.7
Thailand	3.5
Bangladesh	3.5
Bhutan	3.4
Indonesia	3.4
Timor-Leste	3.3
Nepal	3.2
Myanmar	3.2
India	3.1
Afghanistan	3.1
Cambodia	3.1
Burkina Faso	3.0
Sri Lanka	3.0
Rwanda	2.9
Pakistan	2.8

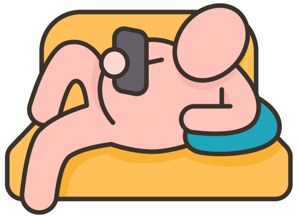
All of the listed countries are LMICs.

Factors Influencing the Obesity Epidemic in LMICs



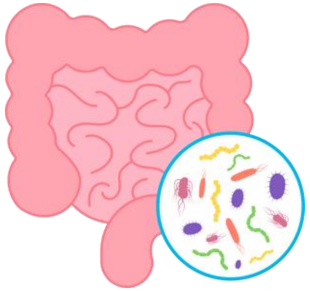
Nutritional Transition

Global Trade



Increased Sedentary Lifestyle

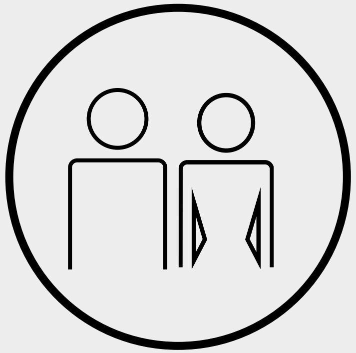
Early Life Undernutrition



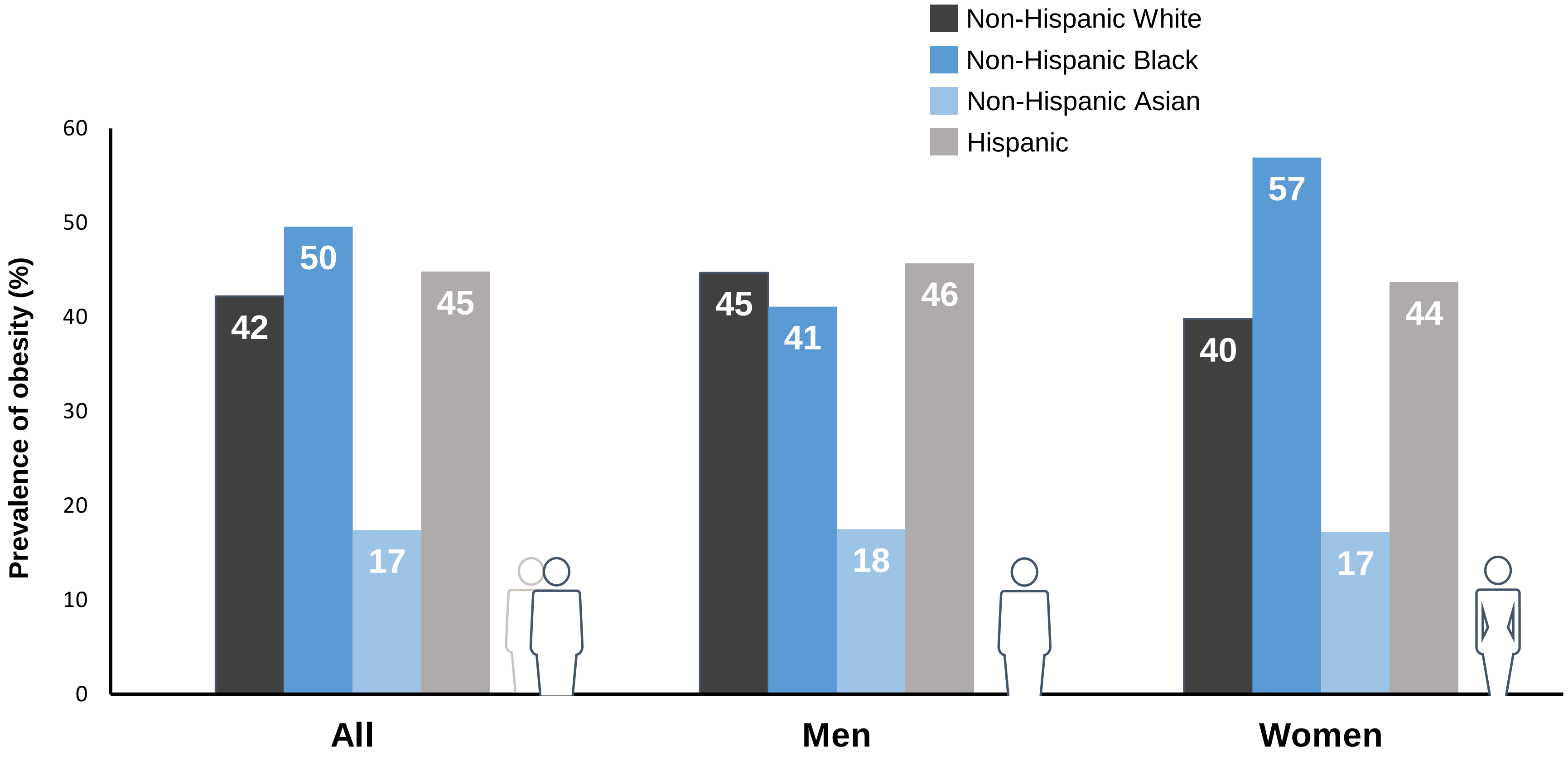
Gut Microbiome

Prevalence of obesity by sex and race

Adults from 2017–2018, U.S.

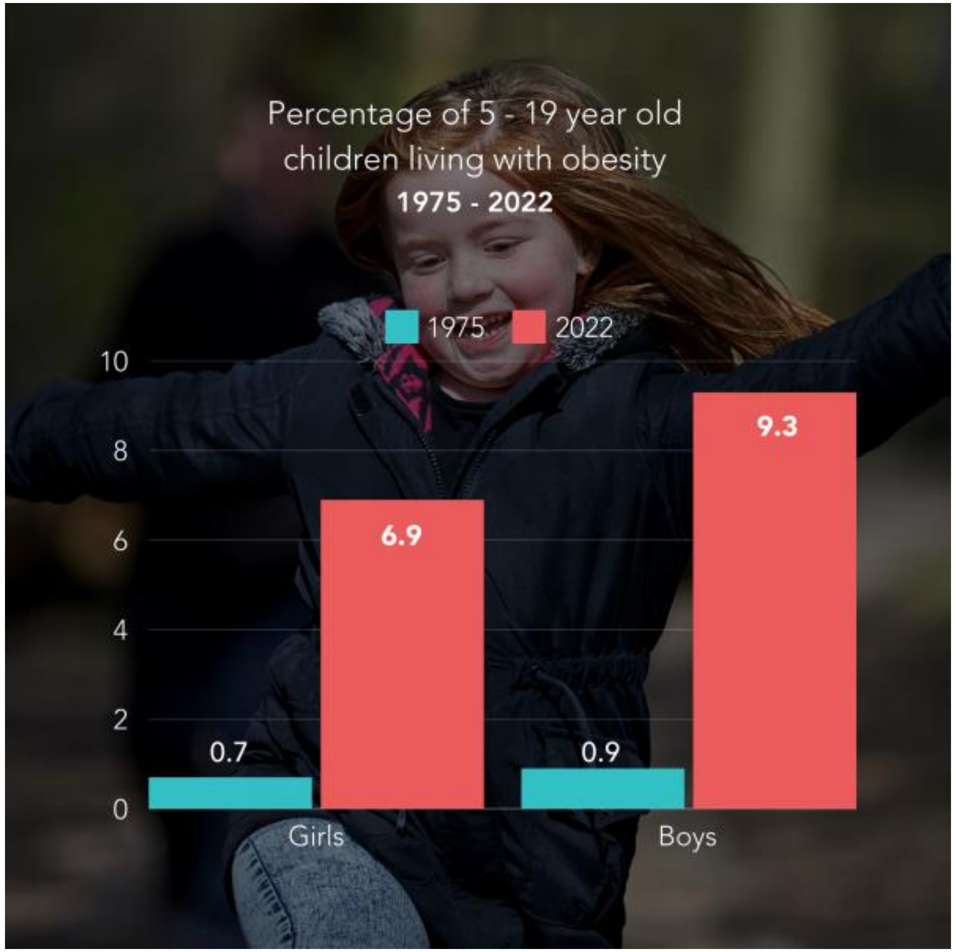


What differences can you see?



Global Trends in Pediatric Obesity

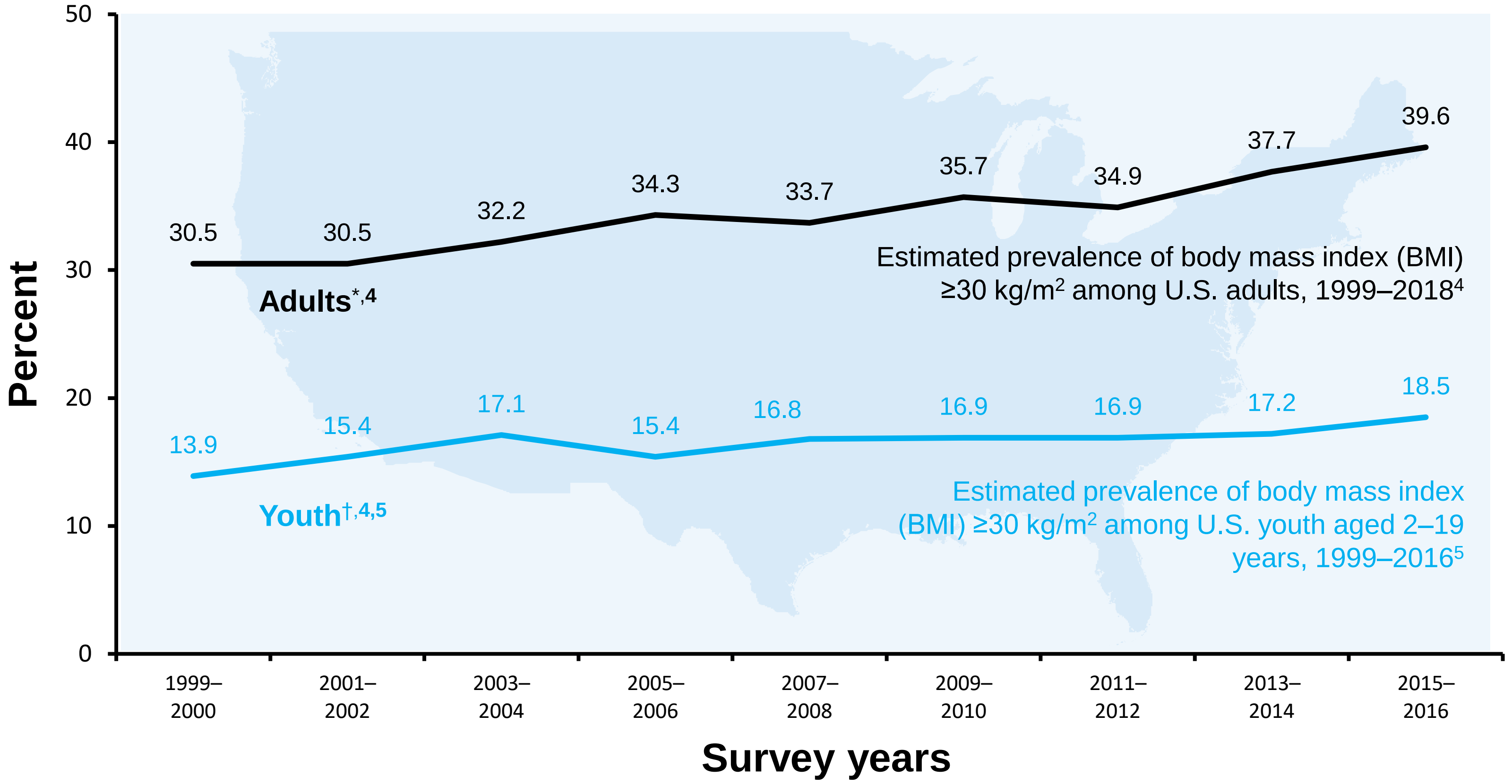
- UNICEF estimates that in 2000, about 30 million children under age 5 were overweight or obese. By 2022, this number increased to 37 million.
- Estimates show the prevalence of overweight and obesity among children aged 5-19 years to have risen from just 4% in 1975 to almost 20% in 2022.
- In 1975, 0.7% of girls were living with obesity compared to 6.9% in 2022 while boys saw an increase from 0.9% to 9.3%.
- Unlike adults, the obesity appears to have a higher prevalence in boys rather than girls.



Obesity prevalence in U.S. over time

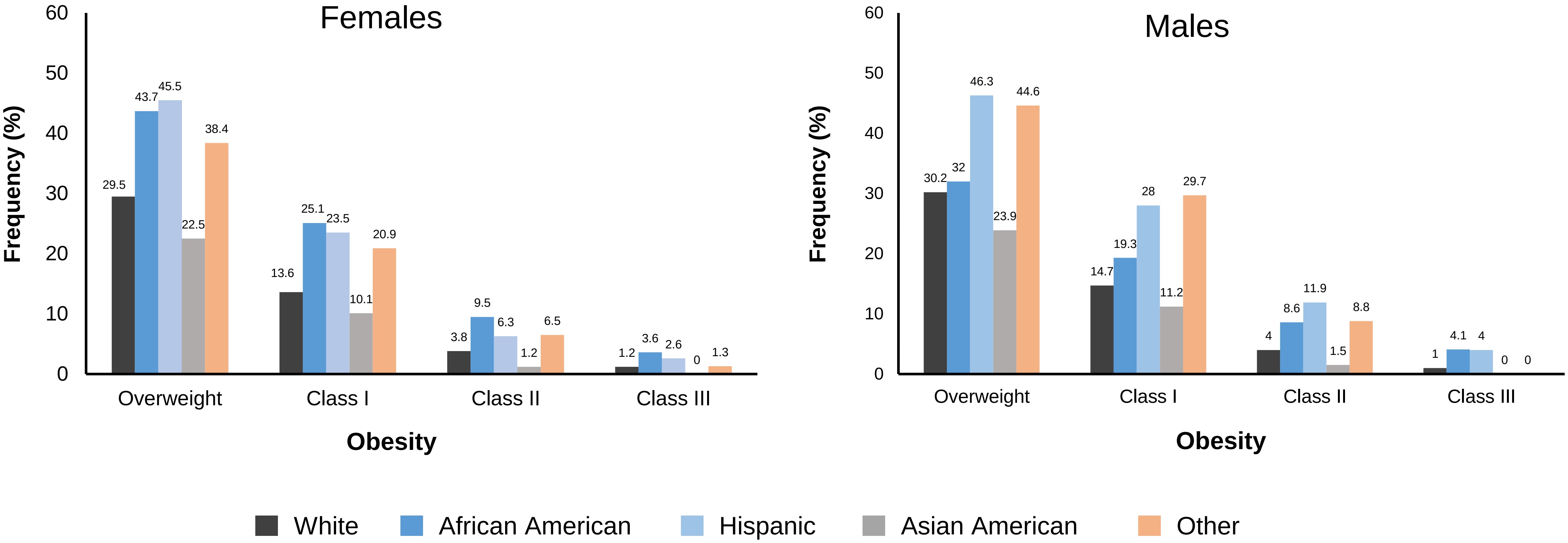
Adults and children

- Pre-obesity and obesity criteria used by the NCHS and the CDC[1]
- Severe obesity classification recommended by the AHA[2,3]



Prevalence of pre-obesity and obesity in U.S. by race

Children from 2015–2016



Skinner AC et al. Pediatrics 2018;141:e20173459.

Presumed Stabilization of Obesity

- Studies analyzing the global prevalence of obesity between 1980 to 2013 have suggested that there is some evidence of stabilization of obesity rates after 2006.
- Pooled analysis of population-based studies which included a total of 20 million participants globally demonstrated BMI increasing slowly in high income countries after 2000.
- However, concurrently, the rate of BMI increase accelerated for men in Central and Eastern Europe and most countries in Latin America and the Caribbean (LAC).
- Evidence suggests that obesity rates, though with high prevalence, are stabilizing for children and adolescents in most economically advanced countries.

1975
13.8 %
Prevalence of underweight in men

2014
8.8%
Prevalence of underweight in men

1975
14.6%
Prevalence of underweight in women

2014
9.7%
Prevalence of underweight in women

What are the proposed theories to support the stabilization of Obesity?

1. **Biological Limit**
   Saturation threshold for the proportion of persons who can become obese, beyond which there is no further increase in prevalence.

2. **Market Penetration**
   The consumption of ultra processed foods in HICs has been consistently elevated for decades and is not projected to increase further.

3. **Effective public health interventions**

4. **Fact or Fiction**

What are the proposed theories to support the stabilization of Obesity?

Despite there being evidence of obesity rates stabilization in high income countries there is a steady increase low to middle income countries, and this has led to a net continued global increase in obesity rates.

Obesity was assessed using BMI which is a crude measure of adiposity and does not capture the true variations in population.

Prevalence trends do not capture incidence nor duration of obesity cases.

What are the proposed theories to support the stabilization of Obesity?

Despite there being evidence of obesity rates **stabilization in high income countries** there is a **steady increase low to middle income countries,** and this has led to a net continued **global increase in obesity rates**.

Obesity was assessed using BMI which is a **crude measure of adiposity** and does not capture the true variations in population.

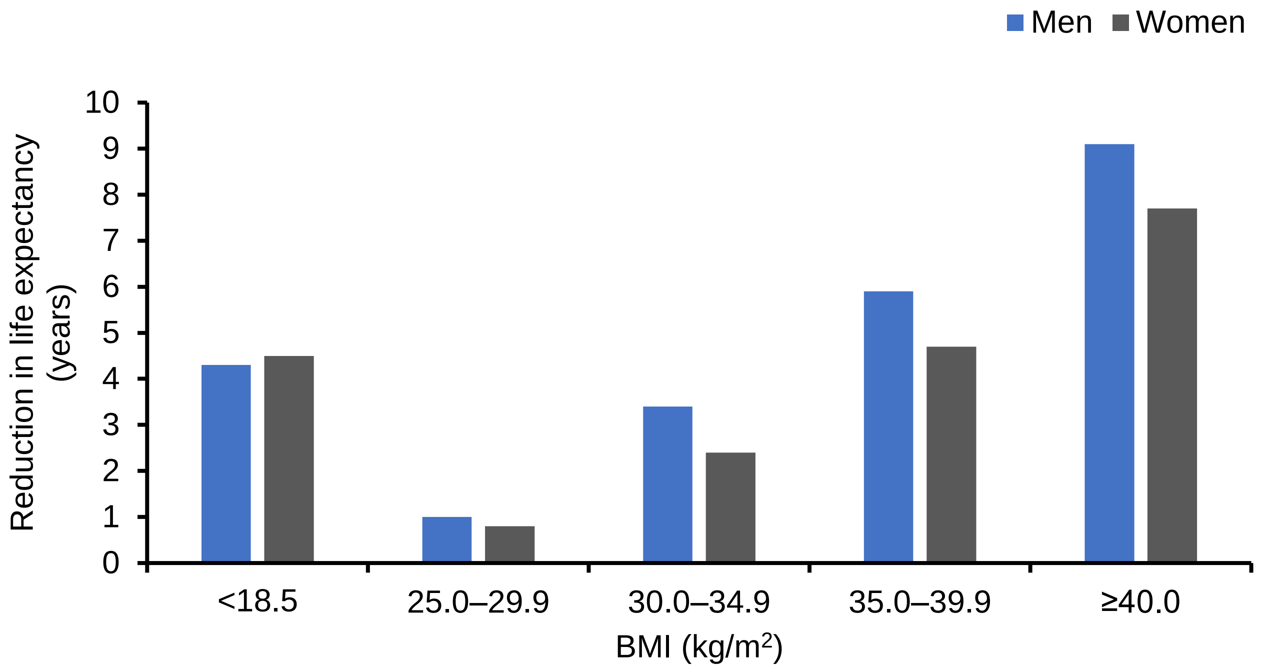
Prevalence trends do not capture **incidence** nor **duration of obesity cases**. Thus, incidence rates could still be increasing during a period of obesity prevalence stabilization if the duration of obesity is reduced due **to improved treatments** or **increased mortality**.

Life expectancy across BMI categories

30% increase in mortality is associated with **every 5 BMI point increase** above a BMI of 25[1]

Underweight was associated with increased mortality, with the strongest association between low BMI and mortality to be in **younger people**[2]

Estimated reduction in life expectancy* compared with an individual of healthy weight (18.5–24.9 kg/m²)
Never-smokers, N = 1,969,648[2]



*Reduction in life expectancy is calculated as expected age of death minus expected age at death in the healthy weight category. Expected age of death at age 40 years estimated from a Poisson model for overall survival with six-category BMI variable, 5-year age bands, sex, and interaction terms for BMI with age at BMI, and BMI with sex.
BMI, body mass index.

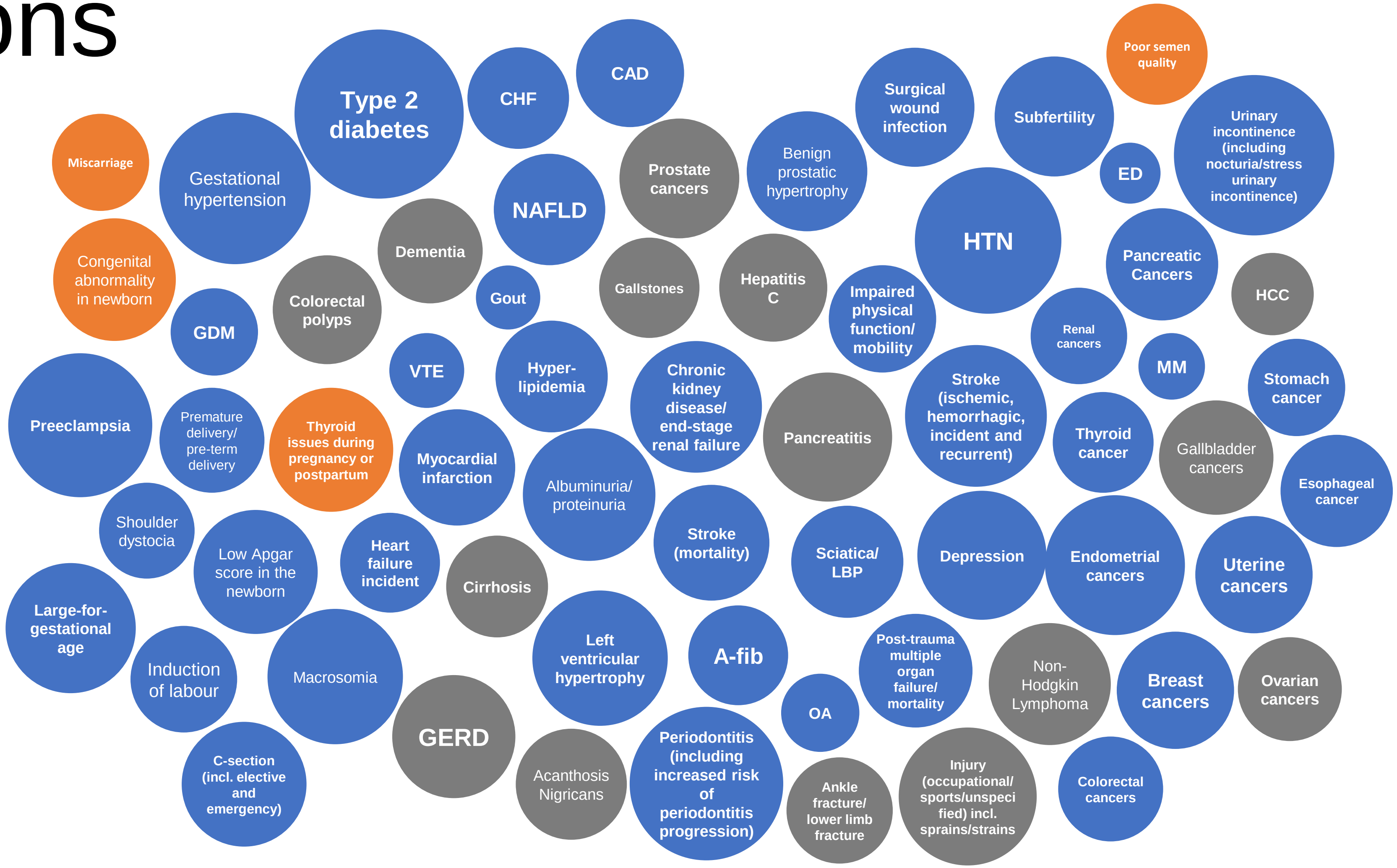
1. Prospective Studies Collaboration. Lancet 2009;373:1083–96; 2. Bhaskaran K et al. Lancet Diabetes Endocrinol 2018;6(12):944–53.

Obesity is associated with multiple complications

	GRADE	Strength of evidence
(blue)	4	Very strong
(grey)	3	Strong
(orange)	2	Moderate
(dark navy)	1	Weak

229+

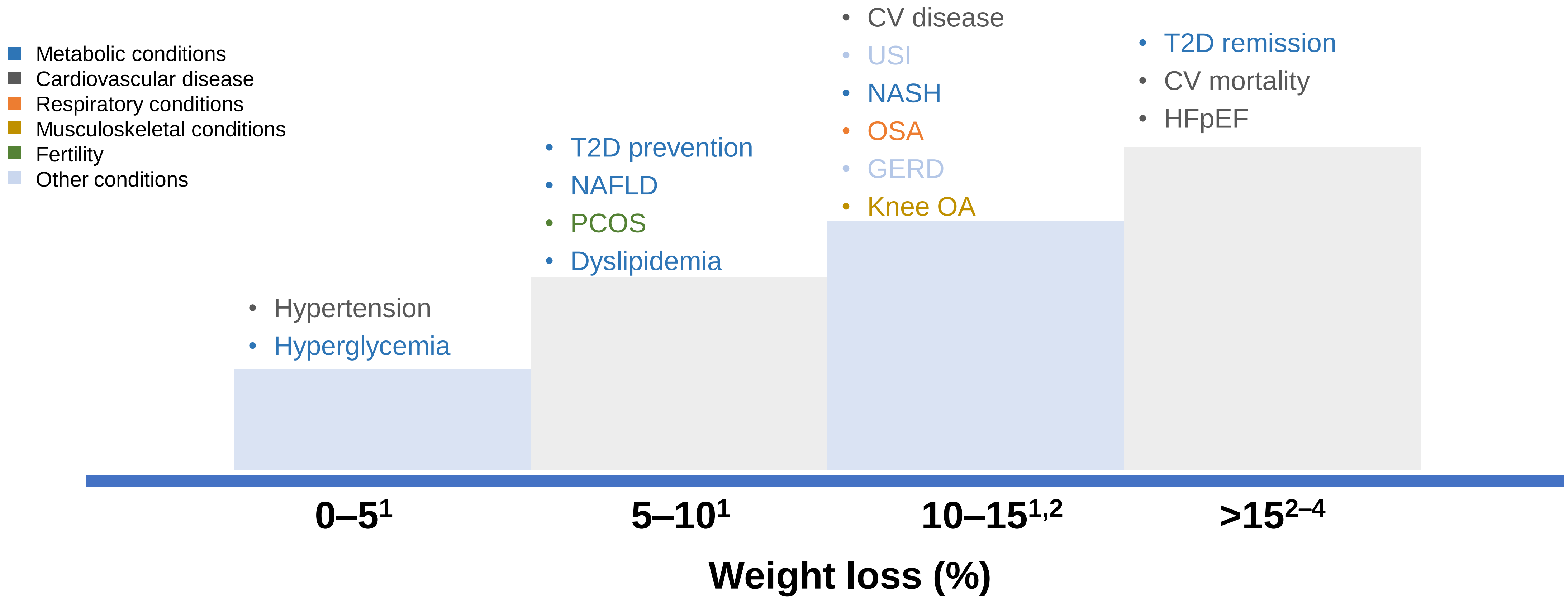
complications affecting *EVERY* organ system and medical specialty



A-fib, atrial fibrillation; CAD, coronary artery disease; CHF, congestive heart failure; ED, erectile dysfunction; GDM, gestational diabetes mellitus; GERD, gastroesophageal reflux disease; HCC, hepatocellular carcinoma; HTN, hypertension; LBP, lower back pain; MM, multiple myeloma; NAFLD, non-alcoholic fatty liver disease; OA, osteoarthritis; VTE, venous thromboembolism

Yuen MM et al. Obesity Week 2016. Oct 31–Nov 4 2016. New Orleans: T-P-3166.

Greater weight loss improves obesity-related complications



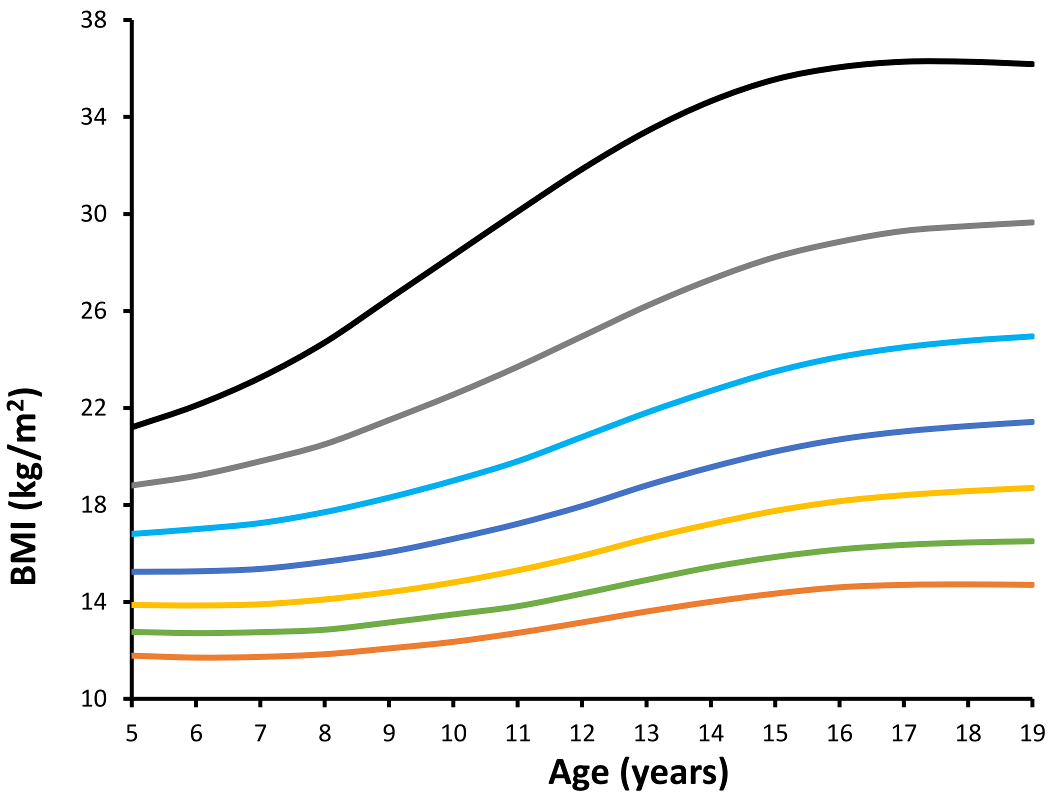
CV, cardiovascular; GERD, gastroesophageal reflux disease; HFpEF, heart failure with preserved ejection fraction; NAFLD, non-alcoholic fatty liver disease; NASH, non-alcoholic steatohepatitis; OA, osteoarthritis; OSA, obstructive sleep apnea; PCOS, polycystic ovary syndrome; T2D, type 2 diabetes; USI, urinary stress incontinence.

1. Garvey WT et al. Endocr Pract 2016;22(Suppl. 3):1–203; 2. Look AHEAD Research Group. Lancet Diabetes Endocrinol 2016;4:913–21; 3. Lean ME et al. Lancet 2018;391:541–51; 4. Benraoune F and Litwin SE. Curr Opin Cardiol 2011;26:555–61.

Defining Overweight/Obesity in Childhood

- **Overweight and obesity are defined statistically by the BMI-for-age or weight-for-recumbent length percentile values that exceed the level considered normal for a child of a specific sex.**
- **BMI z-scores and BMI percentiles can be used to determine cut points and classify weight status of children and adolescents.**
- **A BMI z-score indicates how many standard deviations (SD) the BMI of a child or adolescent is above/below the average BMI for their age and gender.**
- **A positive z-score of 1, 2 or 3 indicates that a child is 1, 2 or 3 SDs above the average value.**

BMI z-score of girls (5–19 years)[2]



Etiology of Childhood Obesity

Obesity in children is multifactorial. Interactions among factors such as genetic predisposition, behavioral and cultural practices, and environmental influences

Genetic factors in pediatric obesity

Syndromic obesity

There are many **single gene** defects associated with obesity have been identified
- Most common is melanocortin-4 receptor (*MC4R*)
- Others include leptin (*LEP*), leptin receptor (*LEPR*), brain-derived neurotrophic factor (*BDNF*) and proopiomelanocortin (*POMC*)

Some **syndromes** associated with obesity include
- Prader-Willi (most common)
- Others include Alström syndrome (AS), Albright hereditary osteodystrophy (AHO), Beckwith-Wiedemann syndrome (BWS)[1], Bardet-Biedl syndrome (BBS) and Cohen syndrome (CS)

Polygenetic obesity is by far the most common genetic cause of obesity
- Accounts for 30–50% of variation in adiposity

Increasing evidence that **epigenetics** play a role
- Epigenetic factors may modify the interaction of environment, microbiome, and nutrition on weight

Environmental causes of obesity in children

Prenatal & parental factors
Smoking during pregnancy, lack of breastfeeding and mimicking parents' poor eating habits are all associated with obesity

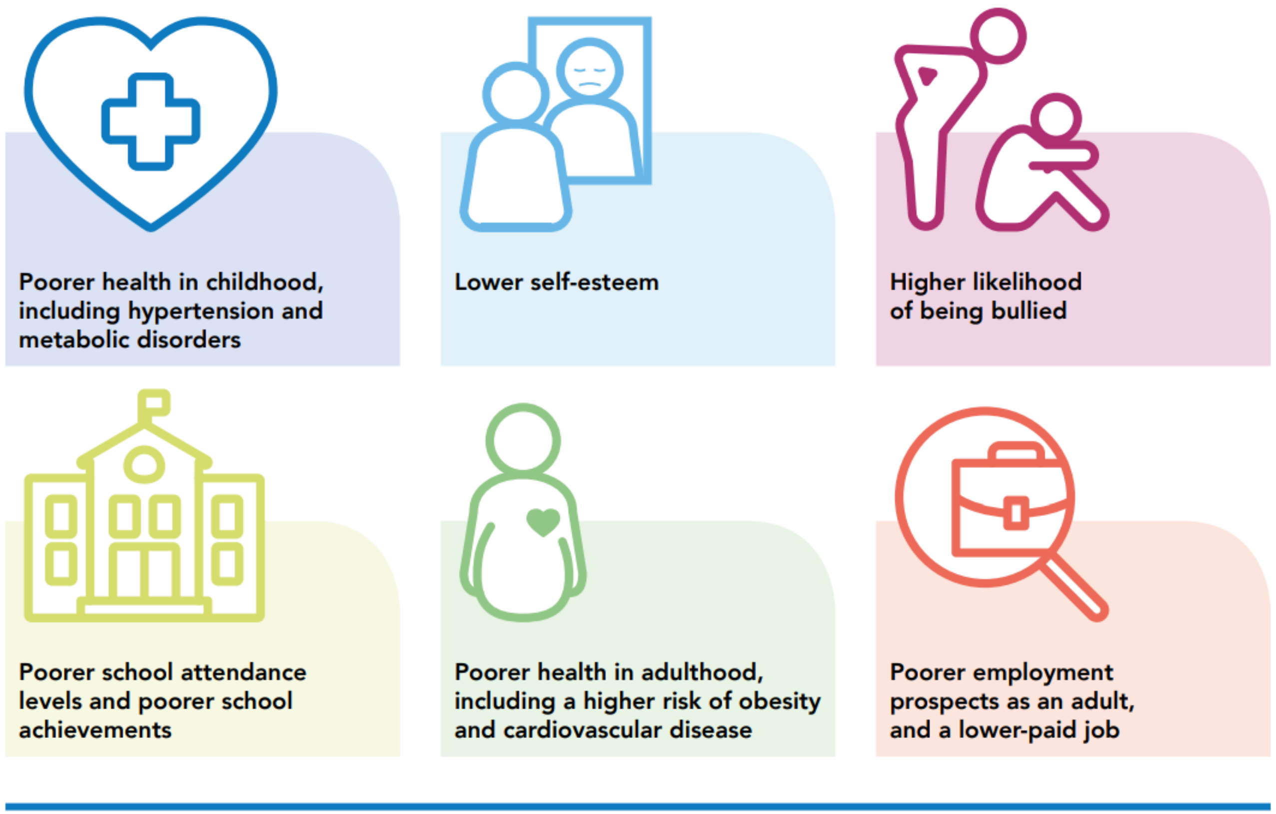
Diet and physical activity
High glycemic foods and sweetened drinks targeted at children. Television, computers, phones, and tablets increased sedentary time.

Socioeconomic factors
Obesity burden greater in lower socioeconomic groups; junk food is inexpensive, and associated with pleasure and convenience

Psychosocial/emotional factors
Depression, peer pressure, anxiety, low self-esteem, body dissatisfaction, eating disorders, poor sleep quality, and maladaptive coping strategies can both cause and result from obesity. Discourage fat-shaming by family members or using shame to motivate the child to lose weight

Environment
Length of the street the children live on, accessibility of nearest playground by foot, frequency of buses/trains passing the street, and the socioeconomic status of the neighborhood

Negative Effects of Childhood Obesity



Obesity can affect a child's immediate health, educational attainment and quality of life.

Children with obesity are very likely to remain obese as adults and are at risk of developing serious noncommunicable diseases.

Thank you!
Any questions?

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